

WBS Roadmap Builder v3.0 – Official User & Technical Guide

Release Date: March 2026

Software Version: 3.0

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PART 1: USER OPERATING GUIDE

1. Introduction

The WBS Roadmap Builder is an integrated Excel-based project management tool. It synchronizes a structured Work Breakdown Structure (WBS) with dynamic Roadmap and Gantt chart visualization engines to provide a unified, automated view of your project's health.

2. The WBS Data Entry Tab

The **WBS Tab** is your command center. Access to this tab is semi-protected to ensure formula integrity while allowing data flexibility.

Work Breakdown Structure				YOUR PROJECT NAME HERE													
Project Start		01-Sep-2025		Update Table													
Current Date		10-Oct-2025		Reset WBS Chart													
				Build Roadmap													
UID	Level	WBS	Task Description	Event Type	Cost	Dependencies (WBSs)	Start	Duration (Days)	Slipped Start	Slipped Duration (Days)	Calculated Start	Calculated End	Calculated Slipped Start	Calculated Slipped End	SDesc	Start Color	End Color
404D-C655	1	1	Project Start	Accomplished Milestone			01-Oct-2025	0			01-Oct-2025	01-Oct-2025			01		
043F-808E	1	2	PCB	Planned Deliverable			20-Sep-2025	0			20-Sep-2025	26-Oct-2025			04		

Core Configuration

- **Project Name:** Located in Column F. This title will appear at the top of all generated charts.
- **Project Start:** All tasks must occur on or after this date.
- **Current Date:** Identifies the "Today" line on your charts. If set earlier than the Project Start, it defaults to the Start Date.

The "Golden Rules" of Data Entry

- **Level Management (Column B):** Use the dropdown to set a task's level (1 to 6). The spreadsheet automatically calculates the WBS number (e.g., 1.2.1) and handles text indentation.
- **Event Types (Column H):** Categorize milestones using the dropdown. These map to specific visual symbols:
 - **△ / ▲** : Milestone (Planned / Accomplished)
 - **◇ / ◆** : Decision Point (Planned / Accomplished)
 - **▽ / ▼** : Deliverable (Planned / Accomplished)

The Automated Calculation Engine (Columns J–N)

These columns are **Read-Only**. The software recalculates these dates based on:

1. **Dependencies:** If Task B depends on Task A, its start date will "slip" to match Task A’s finish.
2. **Parent-Child Rollups:** Top-level tasks (Level 1) automatically adopt the earliest start and latest finish dates of their nested sub-tasks.

3. Generating Your Charts

Click the "**Build Roadmap**" button to trigger the Plot Builder interface.

Plot Builder

Plot Type

☒ Use Cells ☐ Use Shapes

WBS Level

1

Time Select

☒ By Month ☐ By Quarter

Visibility

☐ Don't Plot Hidden Rows

☐ Show Dependency Lines

☐ Show "Loopback" Dependencies

☒ Show Slip Markers

☐ Show Start/Stop Column

Submit Cancel

- **WBS Level:** Choose the depth of the chart. Level 1 shows major phases; Level 3 provides granular detail.
 - **Time Units:** Toggle between **Monthly** or **Quarterly** timeline headers.
 - **Show "Loopback" Dependencies:** This is something that rarely (if ever) happens. When ticked, any item depending on something in its future will get a backwards arrow pointing from the later to the earlier event. With any luck, I've removed this feature, but I kept the tick box just in case the bug isn't fully removed.
 - **Show Start/Stop Column:** Adds two columns with the earliest and latest dates of each WBS item
 - **Show Slipped Markers:** Displays red icons alongside scheduled markers to highlight delays.
 - **Don't Plot Hidden Rows:** If you have collapsed rows in Excel using the grouping (+/-) feature, the builder will ignore them.
 - **Decide if you want the plot to use colored cells or shapes.**
-

PART 2: ADVANCED LOGIC & CUSTOMIZATION

1. Dependency Management

The "Dependencies (WBS#s)" column (Column G) drives the scheduling engine.

- **Multiple Links:** Use comma-separated WBS numbers (e.g., 2.1, 2.2).
- **Constraint:** Only the lowest-level "leaf" tasks should have manual dependencies.
- **Circular References:** If Task A depends on B and B depends on A, the cell will highlight **Red** and the link will be ignored.

2. Visual Coding

- **Slipped Schedule:** Enter values in **Slipped Start** (Column K) and **Slipped Duration** (Column L). This creates a "Ripple Effect," pushing all dependent tasks forward and creating a red "impact" overlay on the charts.
 - **Percent Done (Column N):** Enter 0% to 100%. The cell heat-maps (Red → Yellow → Green) for an instant status check.
 - **Start/End Colors (Columns O-P):** Defines the color and pattern for bars occurring before and after the "Current Date."
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PART 3: TECHNICAL MAINTENANCE (FOR VBA CODERS)

1. System Architecture

The tool utilizes a "Clean Slate" strategy. When a plot is generated, the existing sheet is deleted and a new one is scaffolded from scratch to prevent graphical artifacts.

- **Axis Engine (WbsTimeLineAxis):** Converts calendar dates into Excel X-coordinates.
- **Header Engine (TimelineHeaderBuilder):** Dynamically merges cells to create Fiscal Year (Oct-Sep) and Quarter tiers.
- **Validation (RegExTools):** Uses Regular Expressions to parse WBS strings and dependencies.

2. VBA Reference Requirements

To maintain functionality, the following libraries must be enabled in **Tools > References:**

1. Visual Basic for Applications
2. Microsoft Excel 16.0 Object Library
3. Microsoft Scripting Runtime (For Dictionaries)
4. Microsoft VBScript Regular Expressions 5.5

3. Update Checklist

- **Versioning:** Update VersionNumber.bas before distribution. This stamps the footer of every generated chart.
- **Performance:** To speed up generation for large projects, the UnifiedPlotBuilder suppresses Application.ScreenUpdating and DisplayAlerts during the draw cycle.
- **Protection:** Ensure the WBS tab is protected before deployment to hide calculation logic in Columns Q+.